# DAVID PETRIE

Systems & Security | Distributed Infrastructure | Penn State Class of 2027
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## SUMMARY

Enterprise Technology Integration student at Penn State who builds and runs real systems: a multi-region Solana and Aptos RPC network that peaked at 8M+ transactions in a day, a live SMS product, and automated trading software. Seeking internships and full-time entry-level roles in systems, infrastructure, and security.

## TECHNICAL SKILLS

Programming: Python, Go, TypeScript, JavaScript, Rust, Java, SQL
Infrastructure: Linux, Docker, Caddy, PostgreSQL, TimescaleDB, Azure, Git, CI/CD, Server Orchestration
Security: IDS/SIEM, Snort, Wireshark, Nmap, Nessus, Hydra, Vulnerability Assessment
Blockchain: RPC Design, Solana, Aptos, Yellowstone gRPC, Node Deployment
AI/ML: TensorFlow, PyTorch, Anomaly Detection

## EXPERIENCE

**PixelLabz (formerly PixelPilotz) — Co-Founder & Director of Community Relations | 2022–2025**

- Co-architected a non-consensus RPC network on Solana and Aptos, raising ~$180,000 through token sales.
- Deployed 15+ bare-metal servers in the U.S., Germany, and Singapore; peaked at 8M+ transactions in a day.
- Directed growth to 4,000+ users; delivered demos, onboarding, and technical support.

**NodeStop LLC — Information Security Analyst, Jr (Internship) | 2023–2025**

- Conducted vulnerability assessments and risk analysis for enterprise systems.
- Supported compliance readiness and remediation planning.
- Assisted with IDS/SIEM implementation; communicated findings to technical and non-technical teams.

## PROJECTS & RESEARCH

**Ringlock — Missed-Call Text-Back for Contractors | ringlock.tech | 2026**

- Built and launched a multi-tenant SMS product that texts missed callers back in about 3 seconds.
- Runs on Docker, PostgreSQL, n8n, Twilio, and Caddy; verifies webhook signatures and handles opt-outs.

**sync — Automated Trading System for Kalshi 15-Minute Bitcoin Contracts | 2026**

- Built with Python, FastAPI, React, and TimescaleDB; prices contracts from a synthetic BRTI index.
- Records every trade in a SHA-256 hash-chained audit log; tested on Kalshi's demo environment.

**Kalshi Edge-Finder — Sports Market Pricing Engine in Go | 2026**

- Prices Kalshi sports markets from sharp sportsbook lines and an Elo model, with out-of-sample backtests.
- Keeps real money off until a sport clears backtest and closing-line gates; uses fractional Kelly sizing.

**Machine Learning for Cybersecurity —** Developed active inference models for anomaly detection, optimizing IDS/SIEM efficiency with lower compute overhead.

## EDUCATION

The Pennsylvania State University — B.S. Enterprise Technology Integration | **Expected 2027**
GPA: 3.4
Relevant coursework: Systems Architecture, AI & Machine Learning, Distributed Systems & Blockchain, Data Analytics